United International **University** (**UIU**)
Dept. of Computer Science and Engineering (CSE)
Mid Term Assessment Year: **2025** Semester: **Fall**
Course: **CSE 323/CSE 3711/EEE 4413** Title: **Computer Networks**
Marks: **30** Time: **1 Hour 30 minutes**

[Any examinee found adopting unfair means will be expelled from the trimester/program as per UIU disciplinary rules.]

There are **3 (Three)** questions. Answer **all 3 (Three)** questions. All questions are of values indicated on the right-hand margin.

Q1. a) What are the key components of a data communication system used for networking? Explain with a diagram. **[2+2=4]**

b) Which topology is shown in Fig. 1? Notice that a portion of a cable has been accidentally snapped. Can the stations communicate in this setting? Why or why not? Justify your answer. **[1+2=3]**

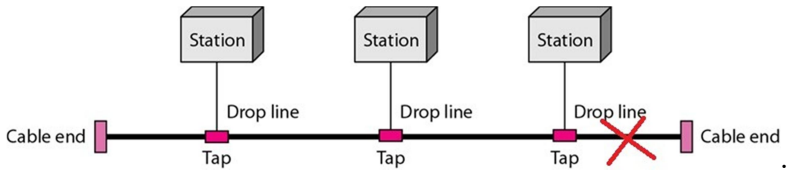
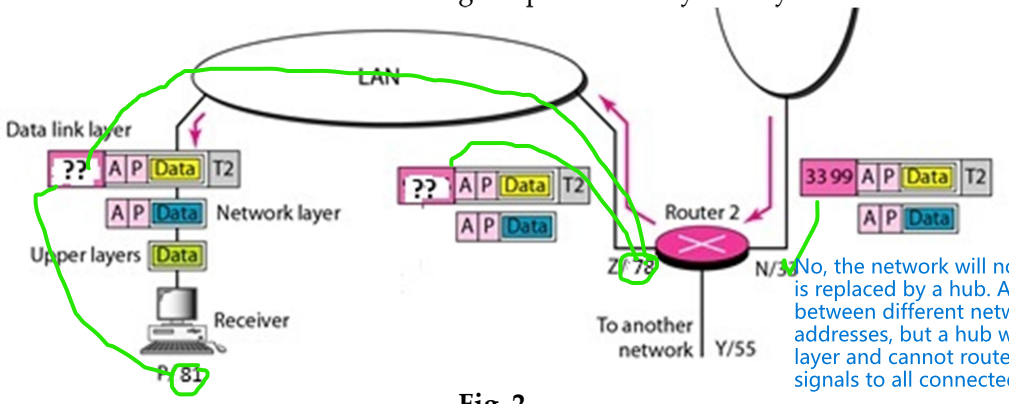


Fig. 1.

c) Carefully look at the diagram in **Fig. 2**. There are two places where **"??"** sign is shown. Determine the values to be used for these two places for the data link layer frames. If the **Router 2** is replaced by a Hub in this scenario, will this network work for forwarding the packets? Why or why not? **[2+1=3]**



No, the network will not work properly if Router 2 is replaced by a hub. A router forwards packets between different networks using logical addresses, but a hub works only at the physical layer and cannot route packets. It only broadcasts signals to all connected devices.

Fig. 2.

Q2. a) What is a **layer** in **OSI reference model**? Why is **layering** used? Why can **layering** be harmful? Explain. **[1+1+1=3]**

b) What is the difference between **logical address** and **physical address**? Which layers of TCP/IP protocol suite are they associated with? Explain clearly with an example diagram. **[1+2=3]**

c) A network communication link has **1 Mbps** link bandwidth. If **59** packets each with **500 bits** are sent with an **average packet arrival rate 2**, what is the ***traffic intensity*** of this network? Show the calculations. Is the traffic intensity of this network link good? Why or why not? How to interpret your result? Justify your answer. **[2+2=4]**

Q3. a) Define a network **socket**. How can a socket be identified for process communications between a sender and a receiver over a network? Explain. **[2+2=4]**

b) Draw a diagram showing the network of networks that the Internet is structured with? Show **IXP** and **peering link** on the diagram. What is the difference between **IXP** and **peering link**? **[3+1+2=6]**

---------- **END of Question** ----------